

AI-Assisted Emergency Triage System Requirements Document

1. Deployment Vision

The AI-Assisted Emergency Triage System will be deployed within Mercer General Hospital's Emergency Department as a web-based clinical decision-support application. The system will integrate with the hospital's Electronic Health Record (EHR) to assist triage nurses during the initial patient assessment.

When a patient arrives, the triage nurse enters or reviews the patient's demographic information, presenting complaints, and vital signs. The AI model analyses the available information and generates a recommended Emergency Severity Index (ESI) level, confidence score, and explanation of the clinical factors influencing the recommendation. The recommendation is displayed on the nurse-facing dashboard, where the nurse can either accept or override the suggested triage level. The final decision remains the responsibility of the clinician.

The system is intended to improve triage consistency, reduce assessment time, and support clinical decision-making without replacing professional judgement.

Human–Machine Interface Requirements

Requirement ID	System Requirement
HMI-01	The system shall require secure login using authorised hospital credentials before any patient information is displayed.
HMI-02	The system shall display the current patient triage queue within 2 seconds of user login.
HMI-03	The system shall display the AI-generated ESI recommendation, confidence score, and explanation within 500 milliseconds after all required patient information has been submitted.
HMI-04	The system shall highlight ESI Level 1 and Level 2 patients using a red visual alert immediately after the AI recommendation is generated.

HMI-05	The system shall allow the triage nurse to accept or override the AI recommendation using a single button selection.
HMI-06	If the AI recommendation is overridden, the system shall require the nurse to enter an override reason before the patient record can be saved.
HMI-07	The system shall automatically save all user actions, timestamps, and final triage decisions within 1 second of confirmation.
HMI-08	The interface shall display all primary clinical information using a minimum 16-point font and high-contrast colours suitable for Emergency Department lighting conditions.
HMI-09	The system shall notify users of missing mandatory patient information before allowing AI analysis to begin.
HMI-10	The system shall automatically log out inactive users after 10 minutes of inactivity to protect patient confidentiality.
HMI-11	The system shall display a confirmation message immediately after patient information has been successfully submitted.
HMI-12	If the hospital network connection is unavailable, the system shall notify the user within 5 seconds and prevent incomplete patient records from being submitted.

Model Inputs

The AI model shall receive:

- Patient demographics (age and sex)
- Presenting complaint
- Heart rate
- Respiratory rate
- Blood pressure
- Body temperature
- Oxygen saturation (SpO₂)
- Pain score
- Available Electronic Health Record (EHR) information
- Nurse-entered clinical observations

Model Outputs

The AI model shall generate:

- Recommended Emergency Severity Index (ESI) level
- Confidence score
- Colour-coded priority alert
- Explanation of the clinical factors influencing the recommendation

Following the recommendation, the triage nurse shall review the information, make the final triage decision, and either accept or override the AI recommendation. All actions shall be recorded in the audit log.

Build Path

Selected Build Path: Web Interface

Justification: The system will be deployed as a secure web-based application because Emergency Department staff already use desktop workstations and tablets connected to the hospital network. A web interface allows centralised deployment, real-time Electronic Health Record integration, simplified maintenance, and immediate access from multiple triage stations without installing software on individual devices. This approach supports the existing clinical workflow while ensuring the AI functions as a decision-support tool under continuous human oversight.